

Part 2: Arguments and Rules of Inference

Objectives

By the end of this lesson, you should be able to:

- Recognise whether a sequence of statements is a valid argument
- Apply rules of inference to check that an argument is valid.

2.1 Valid Arguments in Propositional Logic

An argument is a sequence of statements:

$$\begin{array}{c} p_1 \\ p_2 \\ \vdots \\ p_n \\ \therefore q \end{array}$$

The statements $p_1, p_2, p_3, \dots, p_n$ are called the premises or assumptions, and q is called the conclusion.

The argument is valid if $(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q$ is a tautology. In other words, when *all the premises are true, the conclusion is also true*.

Example 2.1. *Is this a valid argument?*

$$p \rightarrow q$$

$$p$$

$$\therefore q$$

Solution

p	q	$p \rightarrow q$	p	q
T	T	T	T	T
T	F	F	T	F
F	T	T	F	T
F	F	T	F	F

The given argument is valid since when the premises are true, the conclusion is also true.

Example 2.2. *Show that this is an invalid argument.*

If there is fire, then there is smoke.

There is smoke.

Therefore, there is fire.

Solution

Let p denote the statement “there is fire”, and q be the statement “there is smoke”, then the above can be written as

$$p \rightarrow q$$

$$q$$

$$\therefore p$$

Look at the truth table.

p	q	$p \rightarrow q$	q	p
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	F

The given argument is invalid since in the third row the premises are true but the conclusion is false.

2.2 Fallacies

A fallacy is an error that results in an invalid argument. i.e. A fallacy is an argument that is not valid.

2.2.1 Converse Error

A converse error results from an invalid argument of the form:

$$\begin{array}{l} p \rightarrow q \\ q \\ \therefore p \end{array}$$

Example 2.3. *Here is an example of converse error:*

*If I work all night on the homework, then I can answer all the questions.
I can answer all the questions.
Therefore, I worked all night on the homework.*

2.2.2 Inverse Error

A inverse error results from an invalid argument of the form:

$$\begin{array}{l} p \rightarrow q \\ \neg p \\ \therefore \neg q \end{array}$$

Example 2.4. *Here is an example of inverse error:*

*If I have access to the internet, then I'll get an A+.
I do not have access to the internet.
Therefore, I will not get an A+.*

2.3 Rules of Inference

We can apply rules of inference to check that an argument is valid. When an argument becomes too complicated, it is no longer practical to analyse it using a truth table.

Table of Rules of Inference

Modus Ponens	$p \rightarrow q$ p $\therefore q$
Modus Tollens	$p \rightarrow q$ $\neg q$ $\therefore \neg p$
Addition	p $\therefore p \vee q$ q $\therefore p \vee q$
Simplification	$p \wedge q$ $\therefore p$ $p \wedge q$ $\therefore q$
Conjunction	p q $\therefore p \wedge q$
Disjunctive Syllogism	$p \vee q$ $\neg q$ $\therefore p$ $p \vee q$ $\neg p$ $\therefore q$
Hypothetical Syllogism	$p \rightarrow q$ $q \rightarrow r$ $\therefore p \rightarrow r$
Division into Cases	$p \vee q$ $p \rightarrow r$ $q \rightarrow r$ $\therefore r$
Contradiction	$\neg p \rightarrow F$ $\therefore p$
Resolution	$p \vee q$ $\neg p \vee r$ $\therefore q \vee r$

Example 2.5. State which rule of inference is the basis of the following argument: "It is below freezing now. Therefore, it is either below freezing or raining now."

Solution: Let p be the proposition "It is below freezing now" and q the proposition "It is raining now." Then this argument is of the form

$$\begin{array}{l} p \\ \therefore p \vee q \end{array}$$

This is an argument that uses the addition rule.

Example 2.6. State which rule of inference is the basis of the following argument: "It is below freezing and raining now. Therefore, it is below freezing now."

Solution: Let p be the proposition "It is below freezing now," and let q be the proposition "It

is raining now." This argument is of the form

$$\begin{array}{c} p \wedge q \\ \therefore p \end{array}$$

This argument uses the simplification rule.

When there are many premises, several rules of inference are often needed to show that an argument is valid. This is illustrated by Examples 3.5 and 3.6, where the steps of arguments are displayed on separate lines, with the reason for each step explicitly stated. These examples also show how arguments in English can be analysed using rules of inference.

Example 2.7. Show that the hypotheses 'It is not sunny this afternoon and it is colder than yesterday', 'We will go swimming only if it is sunny', 'If we do not go swimming, then we will take a canoe trip', and 'If we take a canoe trip, then we will be home by sunset' lead to conclusion 'We will be home by sunset'.

Solution

Let p be the proposition 'It is sunny this afternoon',
 q the proposition 'It is colder than yesterday',
 r the proposition 'We will go swimming',
 s the proposition 'We will take a canoe trip',
and t the proposition 'We will be home by sunset'.

Then the hypotheses are $\neg p \wedge q$, $r \rightarrow p$, $\neg r \rightarrow s$, and $s \rightarrow t$.

The conclusion is simply t .

We construct an argument to show that our hypotheses lead to the desired conclusion as follows.

Steps	Reason
1. $\neg p \wedge q$	Hypothesis
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Hypothesis
4. $\neg r$	Modus tollens using (3) and (2)
5. $\neg r \rightarrow s$	Hypothesis
6. s	Modus ponens using (5) and (4)
7. $s \rightarrow t$	Hypothesis
8. t	Modus ponens using (7) and (6)

Note that we could have used a truth table to show that whenever each of the four hypotheses is true, the conclusion is also true. However, because we are working with five propositional variables, p , q , r , s and t , such a truth table would have 32 rows.

Example 2.8. Show that the hypotheses ‘If you send me an email message, then I will finish writing the program’, ‘If you do not send me an email message, then I will go to sleep early’, and ‘If I go to sleep early, then I will wake up feeling refreshed’ lead to the conclusion ‘If I do not finish writing the program, then I will wake up feeling refreshed’.

Solution

Let p be the proposition ‘You send me an e-mail message’,
 q the proposition ‘I will finish writing the program’,
 r the proposition ‘I will go to sleep early’, and
 s the proposition ‘I will wake up feeling refreshed’.

Then the hypotheses are $p \rightarrow q$, $\neg p \rightarrow r$ and $r \rightarrow s$.
 The conclusion is $\neg q \rightarrow s$.

We construct an argument to show that our hypotheses lead to the desired conclusion as follows.

Steps	Reason
1. $p \rightarrow q$	Hypothesis
2. $\neg q \rightarrow \neg p$	Contrapositive
3. $\neg p \rightarrow r$	Hypothesis
4. $\neg q \rightarrow r$	Hypothetical syllogism using (2) and (3)
5. $r \rightarrow s$	Hypothesis
6. $\neg q \rightarrow s$	Hypothetical syllogism using (4) and (5)